

Backup Policy

PrimeCore Minerals | Secure Multi Office Network

PC-06 | Version 0.1 | 23 September 2026

Status: Proposed policy for approval

Owner: IT Administrator | Approver: Project lead | Approval date: Pending

Protect PrimeCore configuration, identity, business files and monitoring data with recoverable copies. This policy sets proposed backup requirements that must be proven through restore tests.

Backup operating baseline

Maintain three copies of important data across two failure domains, including one offline or immutable copy with separate administrative credentials. A snapshot on the same EVE host is useful for rollback but does not protect against loss of that host. VLAN 91 is recorded in the addressing register. Later tests on 29–30 September 2026 demonstrate a Restic repository and byte-identical recovery of a Samba AD archive; see the screenshot evidence appendix. This does not establish off-host resilience, approved scheduling or a full domain recovery. [S1]

Data or system	Proposed schedule and retention	Recovery objective
Firewall and network configuration	After approved changes and daily; 30 daily and 12 monthly copies	RPO 24 hours or last approved change; RTO 4 hours
Directory service	Daily supported, consistent backup; 30 daily and 12 monthly copies	RPO 24 hours; RTO 8 hours
Protected shared files	Every 4 hours; retain 7 days of these points, 30 daily and 12 monthly copies	RPO 4 hours; RTO 8 hours
Monitoring configuration	After changes and daily; 30 daily and 12 monthly copies	RPO 24 hours; RTO 8 hours
Security logs and evidence	Daily protected copy; retention and holds under SEC-04	RPO 24 hours for protected copy; RTO 24 hours

RPO is the maximum targeted data loss measured in time. RTO is the target elapsed time to restore service after declared disruption. These values require business-owner approval and measured recovery tests; schedules alone do not prove either target.

Protection requirements

- Encrypt transfers and stored copies where supported. Restrict restore and deletion rights separately, and keep recovery keys available through an authorised emergency process.
- Capture application-consistent backups with the selected product's supported method. Include permissions, ownership and required configuration. Use the directory service's supported recovery method.
- Prevent ordinary endpoints from writing to or deleting backup history. Permit only required backup flows to repository addresses, not unrestricted access to VLAN 91.

Backup operations and assurance

1. The backup operator records job scope, schedule, repository, retention, encryption and owner in the backup register. The data owner confirms protected paths and excluded data.
2. Each working day, check job completion, warnings, copy age, repository capacity and replication or offline-copy status. Log failed jobs as incidents with an owner and next action.
3. For a missed protected-file backup, retry promptly and alert the IT owner when the latest usable point approaches the four-hour RPO. For suspected compromise, protect existing clean copies before reconnecting repositories.
4. Restore selected files monthly and a representative complete service each term or quarterly. Restore to an isolated location, verify content and permissions, then measure recovery time.
5. Before release, the service owner accepts the recovered data. Retain the original affected data until evidence and rollback requirements are satisfied. Record the selected restore point and data lost.

Backup and restore records

Backup job register: job ID; system and paths; data owner; schedule; repository and failure domain; retention; encryption-key custodian; last successful job; oldest and newest restore points; last restore test; exceptions. Restore record: request and approval; incident; restore point; destination; start and finish times; integrity check; ACL test; owner acceptance; data-loss assessment.

Capacity and retention

Size storage from measured protected data, daily change rate, retained recovery points and growth. Monitor both free space and the age of the newest usable backup. SEC-04 proposes 90 days searchable logs, archive to 365 days from event time, and incident evidence until 12 months after closure or longer under a hold. A backup rotation must not delete the only held evidence copy. [S5]

Acceptance criteria

Demonstrate retrieval after a simulated host loss, deny backup deletion by an ordinary user, verify recovery-key access, and restore one file with original permissions. Record actual RTO and RPO. Link exceptions and missed targets to the recovery plan PC-10.

Review and exceptions

The owner reviews this document every six months and after a significant incident or system change. Record exceptions with the affected asset, business reason, risk, compensating control, approver and expiry. Approval does not replace a successful implementation test.

References and linked artefacts

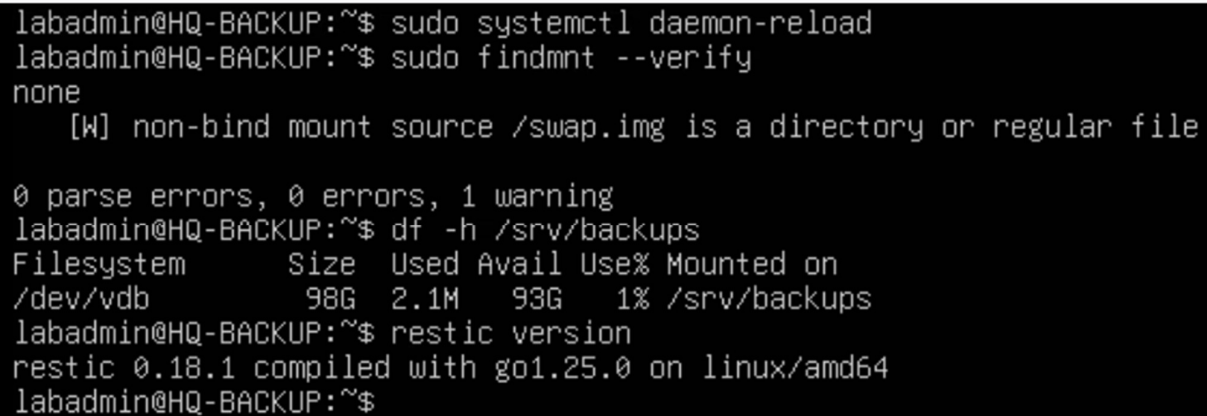
[S1] HQ_Network_Addressing_Artefact.md, project deliverables, evidence dated 13 September 2026.

[S5] SEC-01 to SEC-05 procedures: 01_Network_Access_Procedure.docx; 02_Shared_File_Procedure.docx; 03_Notification_Escalation_Procedure.docx; 04_Logging_Evidence_Procedure.docx; 05_Authentication_Lockout_Procedure.docx. Proposed operating requirements are distinct from implemented controls.

Screenshot evidence appendix

These figures record selected laboratory observations. They support the tested functions described in this artefact. Proposed policies, untested controls and organisational approval remain separate from the pictured results.

Figure E1 Mount validation and Restic version



```
labadmin@HQ-BACKUP:~$ sudo systemctl daemon-reload
labadmin@HQ-BACKUP:~$ sudo findmnt --verify
none
[W] non-bind mount source /swap.img is a directory or regular file

0 parse errors, 0 errors, 1 warning
labadmin@HQ-BACKUP:~$ df -h /srv/backups
Filesystem      Size  Used Avail Use% Mounted on
/dev/vdb         98G   2.1M   93G   1% /srv/backups
labadmin@HQ-BACKUP:~$ restic version
restic 0.18.1 compiled with go1.25.0 on linux/amd64
labadmin@HQ-BACKUP:~$
```

findmnt reports zero errors; its warning is for the existing swap file. vdb is mounted with about 93 GB available; Restic is version 0.18.1.

Source evidence: evidence/Backup_Server/B39_a985fb32-4a28-4a9e-bcb4-92c60d48839a.png

Screenshot evidence continued

Figure E2 Samba online backup completes

```
INFO 2026-09-29 23:19:46,295 pid:93707 /usr/lib/python3/dist-packages/samba/provision/__init__.py #2425: A Kerberos configuration suitable for Samba AD has been generated at /var/backups/primecore-ad/tmp722jy3on/private/krb5.conf
INFO 2026-09-29 23:19:46,295 pid:93707 /usr/lib/python3/dist-packages/samba/provision/__init__.py #2427: Merge the contents of this file with your system krb5.conf or replace it with this one. Do not create a symlink!
Provision OK for domain DN DC=primecore,DC=test
INFO 2026-09-29 23:19:46,311 pid:93707 /usr/lib/python3/dist-packages/samba/join.py #973: Starting replication
Using DS_BIND_GUID_W2K3
Schema-DN [CN=Schema,CN=Configuration,DC=primecore,DC=test] objects[402/1770] linked_values[0/0]
Schema-DN [CN=Schema,CN=Configuration,DC=primecore,DC=test] objects[804/1770] linked_values[0/0]
Schema-DN [CN=Schema,CN=Configuration,DC=primecore,DC=test] objects[1206/1770] linked_values[0/0]
Schema-DN [CN=Schema,CN=Configuration,DC=primecore,DC=test] objects[1608/1770] linked_values[0/0]
Schema-DN [CN=Schema,CN=Configuration,DC=primecore,DC=test] objects[1770/1770] linked_values[0/0]
Analyze and apply schema objects
Partition [CN=Configuration,DC=primecore,DC=test] objects[402/1731] linked_values[0/1]
Partition [CN=Configuration,DC=primecore,DC=test] objects[804/1731] linked_values[0/1]
Partition [CN=Configuration,DC=primecore,DC=test] objects[1206/1731] linked_values[0/1]
Partition [CN=Configuration,DC=primecore,DC=test] objects[1608/1731] linked_values[0/1]
Partition [CN=Configuration,DC=primecore,DC=test] objects[1731/1731] linked_values[54/54]
Replicating critical objects from the base DN of the domain
Partition [DC=primecore,DC=test] objects[97/97] linked_values[23/23]
Partition [DC=primecore,DC=test] objects[288/288] linked_values[24/24]
Done with always replicated NC (base, config, schema)
Replicating DC=DomainDnsZones,DC=primecore,DC=test
Partition [DC=DomainDnsZones,DC=primecore,DC=test] objects[40/40] linked_values[0/0]
Replicating DC=ForestDnsZones,DC=primecore,DC=test
Partition [DC=ForestDnsZones,DC=primecore,DC=test] objects[18/18] linked_values[0/0]
INFO 2026-09-29 23:19:50,122 pid:93707 /usr/lib/python3/dist-packages/samba/join.py #1094: Committing SAM database - this may take some time
Repacking database from v1 to v2 format (first record CN=Print-Start-Time,CN=Schema,CN=Configuration,DC=primecore,DC=test)
Repack: re-packed 10000 records so far
Repacking database from v1 to v2 format (first record CN=IntelliMirrorGroup-Display,CN=C0A,CN=DisplaySpecifiers,CN=Configuration,DC=primecore,DC=test)
Repacking database from v1 to v2 format (first record DC=l.root-servers.net,DC=RootDNSServers,CN=MicrosoftDNS,DC=DomainDnsZones,DC=primecore,DC=test)
Repacking database from v1 to v2 format (first record DC=_ldap._tcp.dc,DC=_msdcs.primcore.test,CN=MicrosoftDNS,DC=ForestDnsZone,DC=primecore,DC=test)
Repacking database from v1 to v2 format (first record CN=Dfs-Configuration,CN=System,DC=primecore,DC=test)
INFO 2026-09-29 23:19:52,348 pid:93707 /usr/lib/python3/dist-packages/samba/join.py #1114: Committed SAM database
INFO 2026-09-29 23:19:52,354 pid:93707 /usr/lib/python3/dist-packages/samba/join.py #1692: Setting isSynchronized and dsServiceName
INFO 2026-09-29 23:19:52,379 pid:93707 /usr/lib/python3/dist-packages/samba/join.py #1656: Cloned domain PRIMECORE (SID S-1-5-21-369086394-685125275-528371242)
INFO 2026-09-29 23:19:52,441 pid:93707 /usr/lib/python3/dist-packages/samba/netcmd/domain/backup.py #290: Backing up sysvol files (via SMB)...
INFO 2026-09-29 23:19:53,021 pid:93707 /usr/lib/python3/dist-packages/samba/netcmd/domain/backup.py #147: Creating backup file /var/backups/primecore-ad/samba-backup-primecore.test-2026-09-29T23-19-52.905930.tar.bz2...
labadmin@hq-dc-01:~$
```

Directory partitions and DNS zones are replicated, SYSVOL is backed up and a primecore.test archive is created. Temporary Kerberos guidance is not an instruction to change the live DC.

Source evidence: evidence/Backup_Server/B45_b527deb9-52d2-4065-9387-cae05f0c3e16.png

Screenshot evidence continued

Figure E3 Active Directory archive recovery

```
labadmin@HQ-BACKUP:~$ sudo mountpoint -q /srv/backups && sudo restic -r /srv/backups/primecore backup /home/labadmin/backup-staging/hq-dc --tag hq-dc-ad
[sudo: authenticate] Password:
enter password for repository:
repository 4104af51 opened (version 2, compression level auto)
no parent snapshot found, will read all files
[0:00] 100.00% 1 / 1 index files loaded

Files:          1 new,      0 changed,      0 unmodified
Dirs:           4 new,      0 changed,      0 unmodified
Added to the repository: 1.396 MiB (1.360 MiB stored)

processed 1 files, 1.394 MiB in 0:07
snapshot e709d068 saved
labadmin@HQ-BACKUP:~$ sudo restic -r /srv/backups/primecore restore latest --tag hq-dc-ad --target /var/tmp/hq-dc-restore-check
enter password for repository:
repository 4104af51 opened (version 2, compression level auto)
[0:00] 100.00% 2 / 2 index files loaded
restoring snapshot e709d068 of [/home/labadmin/backup-staging/hq-dc] at 2026-09-30 04:42:30.016570392 +0000 UTC by root@HQ-BACKUP to /var/tmp/hq-dc-restore-check
Summary: Restored 5 files/dirs (1.394 MiB) in 0:00
labadmin@HQ-BACKUP:~$ sudo diff -s /home/labadmin/backup-staging/hq-dc/hq-dc-ad.tar.bz2 /var/tmp/hq-dc-restore-check/home/labadmin/backup-staging/hq-dc/hq-dc-ad.tar.bz2
Files /home/labadmin/backup-staging/hq-dc/hq-dc-ad.tar.bz2 and /var/tmp/hq-dc-restore-check/home/labadmin/backup-staging/hq-dc/hq-dc-ad.tar.bz2 are identical
labadmin@HQ-BACKUP:~$ _
```

Restic snapshot e709d068 is restored to a separate location and diff reports identical archive bytes. This is file-level archive recovery, not a full restored domain-controller service or a measured RTO/RPO.

Source evidence: evidence/Backup_Server/B51_508a1820-c4d3-4961-8a88-98c48e7e07f2.png